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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Suppose u and v are functions of x that are differentiable at $x = 0$ and that $u(0) = 3$, $u'(0) = -2$, $v(0) = -5$, and $v'(0) = 4$. Find the values of the following derivatives at $x = 0$.

- a. $\frac{d}{dx}(uv)$
 b. $\frac{d}{dx}\left(\frac{u}{v}\right)$
 c. $\frac{d}{dx}\left(\frac{v}{u}\right)$
 d. $\frac{d}{dx}(2v - 7u)$

a. To find $\frac{d}{dx}(uv)$, use the Derivative Product Rule, which states if u and v are differentiable at x , then so is their product uv , and

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}.$$

$$\frac{d}{dx}(uv) = (3)(4) + (-5)(-2) = 22$$

b. To find $\frac{d}{dx}\left(\frac{u}{v}\right)$, use the Derivative Quotient Rule, which states if u and v are differentiable at x and if $v(x) \neq 0$, then the quotient $\frac{u}{v}$ is

$$\text{differentiable at } x, \text{ and } \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}.$$

$$\begin{aligned} \frac{d}{dx}\left(\frac{u}{v}\right) &= \frac{(-5)(-2) - (3)(4)}{(-5)^2} \\ &= -\frac{2}{25} \end{aligned}$$

c. To find $\frac{d}{dx}\left(\frac{v}{u}\right)$, again use the Derivative Quotient Rule.

$$\begin{aligned} \frac{d}{dx}\left(\frac{v}{u}\right) &= \frac{(3)(4) - (-5)(-2)}{(3)^2} \\ &= \frac{2}{9} \end{aligned}$$

d. To find $\frac{d}{dx}(2v - 7u)$, use the Derivative Sum Rule and the Constant Multiple Rule.

The Derivative Sum Rule states that if f and g are differentiable functions of x , then their sum $f + g$ is differentiable at every point where f and g are both differentiable. At such points,

$$\frac{d}{dx}(f + g) = \frac{df}{dx} + \frac{dg}{dx}.$$

The Constant Multiple Rule states that if f is a differentiable function of x , and c is a constant, then $\frac{d}{dx}(cf) = c \frac{df}{dx}$.

$$\begin{aligned} \frac{d}{dx}(2v - 7u) &= 2 \frac{dv}{dx} - 7 \frac{du}{dx} \\ &= (2)(4) - (7)(-2) \\ &= 22 \end{aligned}$$